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SUPERSEDING
Fed. Spec. DD-C-686a
October 10, 1952

FEDERAL SPECIFICATION

CRUCIBLES, FILTERING AND IGNITION; AND COVERS; LABORATORY

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers porcelain ignition and filtering crucibles and crucible covers.

1.2 Classification.

1.2.1 *Crucibles.* Crucibles shall be of the following types and sizes, as specified:

Type I—Ignition crucibles.

Size 1—5 milliliters (ml.).
Size 2—10 ml.
Size 3—15 ml.
Size 4—30 ml.
Size 5—50 ml.
Size 6—100 ml.
Size 7—250 ml.

Type II—Filtering crucibles.

Size 1—1.5 ml.
Size 2—13 ml.
Size 3—25 ml.
Size 4—40 ml.

1.2.2 *Crucible covers.* Crucible covers shall be of the folowing sizes, as specified:

Size 1—17 millimeters (mm.)
Size 2—26 mm.
Size 3—34 mm.
Size 4—38 mm.
Size 5—44 mm.
Size 6—52 mm.
Size 7—66 mm.
Size 8—77 mm.
Size 9—105 mm.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

Federal Specifications:

UU-P-268—Paper, Kraft, Untreated, Wrapping.
UU-T-111—Tape; Paper, Gummed (Sealing and Securing).
PPP-B-576—Box, Wood, Cleated, Veneer, Paper Overlaid.
PPP-B-591—Boxes, Fiberboard, Wood-Cleated.
PPP-B-601—Boxes, Wood, Cleated-Plywood.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636—Box, Fiberboard.
PPP-C-843—Cushioning Material, Cellulosic.

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Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specification:

MIL–L–10547—Liners, Case, Waterproof.

Military Standards:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications, Standards, and Drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following publications form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Uniform Freight Classification Rules.

(Application for copies should be addressed to Uniform Classification Committee, 202 Union Station, Chicago 6, Ill.)

American Society for Testing Materials:

ASTM Standard C242–58T—Terms Relating to Ceramic Whitewares and Related Products.

(Applications for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pa.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Porcelain.* The crucibles and crucible covers shall be a high quality vitreous ceramic whiteware having an average thermal coefficient of linear expansion not greater than 3.6×10^{-6} centimeter (cm)/cm/1°C. between 20° and 200°C. and gradually increasing to not greater than 4.7×10^{-6} cm/cm/1°C. at 1000°.

3.1.1.1 *Thermal shock resistance.* No cracking shall occur when crucible and crucible covers are tested as specified in 4.6.1.

3.1.1.2 *Glaze.* The glaze shall offer a high degree of inertness and have a thermal coefficient of linear expansion approximately equal to that of the porcelain body. Crucibles shall be glazed inside and outside with the exception of the outside bottom surface. Covers shall be glazed inside and outside with the exception of the bottom edge.

3.1.1.2.1 *Glaze adhesion.* Permanent adhesion of the glaze to another glazed porcelain surface of the same material shall not occur below 1250°C. when tested as specified in 4.6.2.

3.1.1.2.1.1 *Permanent adhesion.* Adhesion shall be considered permanent when, after cooling to room temperature, the touching surfaces are annealed as one, or the touching surfaces separate with a portion of one surface adhering to and becoming part of the other surface.

3.1.1.2.2 *Resistance to chemical attack.* Resistance to chemical attack shall be measured by loss of weight by boiling in a solution of

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sodium hydroxide. Weight loss shall be not more than 0.144 percent when tested as specified in 4.6.3.

3.1.1.2.3 *Solubility.* No watermark shall remain on the inner surface of the crucible and no pink color shall develop in the test solution when tested as specified in 4.6.4.

3.2 Design.

3.2.1 *Shape.*

3.2.1.1 *Crucibles.* Type I crucible shall have a flat base to give a firm and stable resting surface. The side wall shall form a sloping and convex surface extending up to the brim. Type II crucibles shall be the same as type I except the base shall be perforated and size 1 shall have straight, tapered sides.

3.2.1.2 *Crucible covers.* The crucible covers shall be designed for use with the respective sizes of crucibles (see tables I and II). There shall be a ring molded in the center of the top of the crucible cover perpendicularly to serve as a handle.

3.2.2 *Dimensions and capacities.*

3.2.2.1 *Crucibles, type I.* The dimensions and capacities shall be as shown in table I.

3.2.2.2 *Crucibles, type II.* The dimensions and capacities shall be as shown in table II.

3.2.2.3 *Crucible covers.* The dimensions shall be as shown in tables I and II. In addition, all crucible covers, of a particular size, shall fit all crucibles of the appropriate size without binding when tested as specified in 4.6.5.

TABLE I.—*Dimensions of type I crucibles and crucible covers*

	Crucibles				Covers	
Size	Capacity* ml.	Top outside diameter** mm.	Base outside diameter** mm.	Height** mm.	Inside diameter** mm.	Size
1	5	24	12	20	26	2
2	10	30	15	26	34	3
3	15	35	18	29	38	4
4	30	50	18	31	52	6
5	50	58	22	36	66	7
6	100	73	27	45	77	8
7	250	99	37	61	105	9

* Tolerance + 15 percent. ** Tolerance ± 1 mm.

TABLE II.—*Dimensions of type II crucibles and crucible covers*

	Crucibles				Perforations		Covers	
Size	Capacity* ml.	Height mm.**	Top outside diameter mm.**	Base outside diameter mm.**	Each diameter mm.***	Area diameter mm.**	Inside diameter mm.**	Size
1	1.5	15	15	12	0.5	7.5	17	1
2	13	34	29	18	0.7	11.0	34	3
3	25	41	36	22	0.7	16.0	38	4
4	40	47	40	25	0.6	19.0	44	5

* Tolerance + 15 percent. ** Tolerance ± 1 mm. *** Tolerance ± 0.02 mm.

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3.3 Identification marking. Each crucible and crucible cover shall be permanently and legibly marked with the name or trademark of the manufacturer. Markings shall be legible after shock, adhesion, and chemical tests have been performed.

3.4 Workmanship. Crucibles and crucible covers shall be symmetrical in shape, uniform in size, and free from cracks, crazing, and other irregularities or foreign matter which detract from their appearance or which may impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Special provisions.

4.2.1 *Alternative inspection (including testing) procedures.* The supplier may utilize any alternative inspection procedure which will insure equal or better quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 *Supplemental evidence.* The supplier shall provide evidence acceptable to the contracting officer that the requirements of 3.1.1, 3.1.1.2, and section 5 have been satisfied.

4.3 Lotting. A lot of crucibles or crucible covers shall consist of those crucibles of the same type and size, or covers of the same size, made by one manufacturer and submitted for acceptance at one time.

4.4 Sampling.

4.4.1 *For nondestructive examination.* Sampling shall be conducted in accordance with Military Standard MIL–STD–105.

4.4.2 *For test.* Fourteen crucibles or crucible covers shall be taken at random from each lot in accordance with the number of specimens required for test as shown in table III.

TABLE III.—*Number of random specimens required*

Test	Number of specimens	Reference paragraph
Thermal shock resistance ..	4	4.6.1
Glaze adhesion	4	4.6.2
Resistance to chemical attack	3	4.6.3
Solubility	3	4.6.4
Binding	Total sample*	4.6.5

* Each crucible cover in the sample shall be tested as specified prior to testings as specified in 4.6.1, 4.6.2, 4.6.3, and 4.6.4.

4.5 Examination. Crucibles and crucible covers and the level A and level B preparation for delivery thereof shall be examined in accordance with the classification of defects and with Military Standard MIL–STD–105. Definitions of porcelain defects shall be in accordance with ASTM Designation C242–58T.

4.5.1 *Classification of defects.*

4.5.1.1 *Crucibles and crucible covers.*

Categories	Defects
Critical:	None defined
Major:	*AQL 4.0 percent defective*
101	Capacity/dimensions incorrect
102	Damage (cracks, crazing, chipping, pinholes, blistering, crawling, eggshelling, orange peel, or shivering)
103	Contamination (grease, oil, or foreign matter)
104	Markings missing, incorrect, or illegible

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4.5.1.2 *Levels A and B preparation for delivery* (section 5).

Categories	Defects
Critical:	None defined
Major:	*AQL 1.5 percent defective*
101	Wrapping or intermediate or shipping container not as specified
102	Number of crucibles or crucible covers per unit or intermediate container not as specified
103	Unlike crucibles (type or size) or crucible covers (size) packaged per unit, intermediate, or shipping container not as specified
104	Closure of unit, intermediate, or shipping container not as specified
105	Unit, intermediate, or shipping container damaged
106	Cushioning material not as specified
107	Marking of intermediate or shipping container illegible, incorrect, or incomplete

4.6 Tests. Distilled water and analytical reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.6.1 *Thermal shock resistance.* The crucible or crucible cover shall be heated to 160° ± 2°C. in an oven for 30 minutes, removed, and immediately plunged into water at a temperature of 9° ± 1°C.

4.6.2 *Glaze adhesion.* Place two crucibles or crucible covers so that the glazed surfaces are in contact with each other and heated to 1250°C. in an electric muffle-type furnace for 30 minutes. Allow to cool by radiation to room temperature.

4.6.3 *Resistance to chemical attack.* Tared crucibles or crucible covers shall be boiled in 10 percent solution of sodium hydroxide for 8 hours, rinsed, dried, and reweighed. Weight loss shall be calculated as follows:

$$L = \frac{100\ (B-A)}{B}$$

Where: L = percent weight loss
B = weight, before boiling, in grams
A = weight, after boiling, in grams

4.6.4 *Solubility.* Each crucible or crucible cover, taken for test, shall be cleaned and rinsed three times with distilled water, using a new quantity of distilled water for each rinsing. The crucibles or crucible covers shall be air-dried in an inverted position for 1 minute after rinsing. After the final drying operation, each type I crucible shall be half-filled with distilled water at 60°C. containing 0.2 percent phenolphthalein solution (prepared from 1 gram of phenolphthalein in 100 ml. alcohol) and autoclaved under 15 pounds steam pressure for 30 minutes. Crucible covers and type II crucibles shall be immersed individually in distilled water containing 0.2 percent phenolphthalein solution in a beaker of suitable size; the crucible covers upside down and the type II crucibles in an upright position. The distilled water used in this test shall have just been boiled for 10 minutes and allowed to cool to 60°C., prior to the addition of the phenolphthalein solution.

NOTE: Glassware used in this test shall not affect test solution.

4.6.5 *Binding.* Each crucible cover, in the sample, shall be rotated 360°, in place, on an accepted crucible of appropriate size.

4.7 Acceptance/rejection criteria. If any crucible or crucible cover fails to meet the test requirements of the specification, the lot represented by the crucible or crucible cover shall be rejected.

5. PREPARATION FOR DELIVERY

For civil agency procurement, the definitions and applications of the levels of packaging and packing shall be in accordance with Federal Standard No. 102.

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5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 *Level A.*

5.1.1.1 *Crucibles.* Each crucible shall be wrapped in 30-lb basis weight kraft paper conforming to grade B of Federal Specification UU–P–268. The wrapping shall be secured with tape conforming to Federal Specification UU–T–111.

5.1.1.2 *Crucible covers.* Each crucible cover shall be wrapped in kraft paper and secured as specified in 5.1.1.1.

5.1.1.3 *Intermediate packaging.* Twelve crucibles of the same type and same size packaged in accordance with 5.1.1.1 or twelve crucible covers of the same size packaged in accordance with 5.1.1.2 shall be packaged in a fiber box conforming to type I, class 1, style optional of Federal Specification PPP–B–636. The box shall be provided with full height partitions forming a cell for each item or contain sufficient cushioning material conforming to type II, class B of Federal Specification PPP–C–843 to prevent shifting of the items. The box shall be closed in accordance with the appendix of the box specification.

5.1.2 *Level C.* Crucibles or crucible covers shall be packaged in uniform quantities of like items in accordance with manufacturer's commercial practices. Each package shall contain only one size and one type of crucible or one size crucible cover. Packages of like items shall be of uniform size, shape, and material.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 *Level A.* Crucibles of the same type and same size or covers of the same size packaged in accordance with 5.1.1.3 shall be packed in a shipping container conforming to Federal Specification PPP–B–636, class 2, V3c or V3s; PPP–B–621, class 2; PPP–B–601, overseas type; PPP–B–591, overseas type; or PPP–B–576, overseas type. As far as practical, exterior containers shall be of uniform size and shape, be of minimum cube and tare consistent with the protection required, and contain identical quantities of like items. The gross weight of each pack shall be limited to approximately 200 pounds. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto. Containers shall be provided with a case liner conforming to Military Specification MIL–L–10547 which shall be sealed in accordance with the appendix thereto.

5.2.2 *Level B.* Crucibles or crucible covers shall be packed in accordance with 5.2.1 except that shipping containers shall comply with Federal Specification PPP–B–636, class 2, W5c or W5s; PPP–B–621, class 1; PPP–B–601, domestic type; PPP–B–591, domestic type; or PPP–B–576, domestic type. Case liners shall not be required.

5.2.3 *Level C.* Crucibles or crucible covers shall be packed in uniform quantities in such a manner as to insure carrier acceptance and safe delivery at first destination for immediate use. Containers shall be in accordance with Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation. Containers for like items shall be of uniform size, shape, and material.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, marking shall be in accordance with Federal Standard No. 123.

5.3.2 *Military agencies.* In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 Intended use.

6.1.1 *Type I.* Type I crucibles are intended for use in ignition of chemical substances for ash content.

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6.1.2 *Type II.* Type II crucibles are intended for use in filtration.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Level of packaging and packing required.

(c) Type and size of crucible required.

(d) Size of crucible cover required.

6.3 Military use. Items procured under this specification for military use are to be limited to the variety shown on the applicable military standards. Personnel of the military departments are requested to refer to these documents for guidance.

6.4 Supersession data. Type A, plain crucibles, and type B, Gooch crucibles, in Federal Specification DD–C–686a have been replaced by type I, ignition crucibles, and type II, filtering crucibles, respectively, in this specification. Type A, Federal Specification DD–C–686a, covered a high form type having a capacity of 30 milliliters. The 30-milliliter capacity type I crucible of this specification is a wide form type. The item covered by type B of Federal Specification DD–C–686a is included in the type II crucibles of this specification.

6.5 Transportation descriptions. Transportation descriptions applicable to these items are:

Crucibles with or without covers.

Crucibles, not otherwise indexed by name.
Carload minimum weight 24,000 pounds, subject to Rule 34, Uniform Freight Classification.

Crucibles, not otherwise indexed.
Motor volume minimum weight 24,000 pounds.

Covers.

Porcelainware, not otherwise indexed by name.
Carload minimum weight 24,000 pounds, subject to Rule 34, Uniform Freight Classification.

Porcelainware, not otherwise indexed.
Motor volume minimum weight 24,000 pounds.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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